

From Disclosure to Topology: A Temporal Multiplex Framework for Extracting Firm Interdependence from 10-K Filings

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Motivation. Firms are embedded in overlapping systems of rivalry, product substitution, and shared exposure, yet text-based firm networks in finance are typically single-layer similarity graphs (TNIC [5]) or firm-level scores [6]. SEC Form 10-K filings are an annually refreshed, mandated corpus in which firms self-report *distinct* relation types in distinct sections. We treat the 10-K as a generative substrate for a node-aligned *temporal multiplex* [1,2,4] and ask: **how do layer-specific topology and interlayer coupling co-evolve as the information environment shifts?** We contribute (i) a schema mapping 10-K Items onto node-aligned layers with financial node attributes, (ii) explicit edge-formation rules, including rules for undisclosed competitors and asynchronous fiscal calendars, and (iii) a hand-curated pilot on Alphabet (G), Meta (M), and Nvidia (N), FY2021–2025, that surfaces the pathologies an automated pipeline must resolve.

Positioning. Three literatures meet here. Finance and accounting extract relations from filings but collapse them into one layer: TNIC [5] links firms by cosine similarity of Item 1 vocabularies, competition scores count rivalry words [6], and customer–supplier links come from major-customer notes [8]. Network science supplies the multilayer formalism [1,2,4] and multiplex diagnostics such as edge overlap and interlayer degree correlation [3], but is rarely applied to regulatory text, where the extraction step itself determines topology. Our aim is to make that step explicit: every edge carries its source Item, its formation rule, and its alignment convention, so that structural claims can be audited against the disclosure that produced them.

Multiplex construction. Let $\mathcal{M}_t = \{G_t^{(\alpha)} = (V, E_t^{(\alpha)}, \mathbf{x}_t)\}_{\alpha}$ share node set V and node attributes $\mathbf{x}_{i,t}$ drawn from Item 8 segment notes: total revenue, segment revenue-share vector $\mathbf{s}_i$, long-lived-asset geography, and major-customer concentration. **L1 Product** (Items 1, 7): each firm maps to a set C_i over a 9-category taxonomy (advertising, devices, XR, cloud/data-center compute, AI platforms, content & streaming, enterprise software, autonomous mobility, health) with breadth weight $J_{ij} = |C_i \cap C_j|/|C_i \cup C_j|$; mapping segment revenues onto the same taxonomy gives a depth weight $R_{ij} = \sum_k \min(s_{ik}, s_{jk}) / \sum_k \max(s_{ik}, s_{jk})$. **L2 Competition** (Item 1): arc $i \rightarrow j$ if i names j ; a *proxy-injected* arc is added when i lists a competitor category that j ’s L1 set satisfies. **L3 Risk** (Item 1A): a bipartite firm–risk-theme graph (21 coded themes) projected with Jaccard weights. A sentiment layer (Item 7 embeddings) is deferred. Layers differ in directionality and edge semantics; we summarize $\mathcal{M}_t$ with the edge overlap $o_{ij} = \sum_{\alpha} a_{ij}^{(\alpha)}$, the fraction of dyads active in ≥ 2 layers, and Spearman correlation of node strength across layers [3]. Temporal change is measured per layer as $\Delta_t^{(\alpha)} = 1 - \text{sim}(E_t^{(\alpha)}, E_{t-1}^{(\alpha)})$, separating edge turnover from weight drift.

Table 1: Item 8 node attributes at the ends of the period-aligned window (Nvidia FY $t+1$).

Firm	Revenue (\$B)		Largest segment share		US share of long-lived assets	
	2021	2024	2021	2024	2021	2024
Alphabet	257.6	350.0	.92 (Services)	.87 (Services)	.73	.75
Meta	117.9	164.5	.98 (Family of Apps)	.99 (Family of Apps)	.79	.86
Nvidia	26.9	130.5	.46 (Gaming)	.88 (Data Center)	.73	.58

Nvidia’s attribute vector moves most: its dominant segment flips from Gaming to Data Center, revenue grows almost fivefold, its asset base shifts toward Taiwan and Israel, and customer concentration rises from one to three $\geq 10\%$ customers (FY2024→FY2025). Treating these as attributes rather than a fifth similarity layer avoids double-counting product information already in L1 and lets attributes condition edges, e.g., testing whether L2 proxy arcs form toward nodes whose segment share in the named category is rising.

Pilot data and coding protocol. For each firm-year we read Items 1, 1A, 7, and 8 of the 10-K (15 filings) and recorded three hierarchical outlines: product lines (segment $\rightarrow$ offering $\rightarrow$ named product), competitor disclosures (category $\rightarrow$ named firms), and risk factors (section heading $\rightarrow$ risk title). Segment revenue, long-lived-asset geography, and major-customer notes were transcribed from the segment note of the financial statements. Offerings were mapped to taxonomy categories by a written rule: a category is assigned only when the filing names a revenue-generating offering in it, not when a technology is merely discussed. Risk titles were coded to themes only when a stand-alone risk factor addresses the theme, so a passing mention inside another factor does not create an edge. The proxy rule for L2 is likewise mechanical: arc $i \rightarrow j$ is injected iff a competitor category listed by i in year t matches a category in $C_{j,t}$ under the same mapping table, which makes injected arcs reproducible and separately reportable.

Asynchronous clocks and structural breaks. Nvidia’s fiscal year ends in late January, so FY t covers roughly calendar $t-1$, lagging G and M by ≈ 11 months. Following the greatest-calendar-overlap convention, we *period-align* Nvidia FY $t+1$ with G/M FY t (common window 2021–2024) and treat filing-label alignment as a robustness check. Financial attributes also carry disclosure breaks that must be modeled, not ignored: Alphabet’s 2023 segment recast (DeepMind to Alphabet-level costs), Nvidia’s Data Center split into Compute and Networking, and ASU 2023-07 segment-expense reporting from FY2024.

Extraction pathologies. (a) *Naming collapse*: in 2021 Alphabet named 24 rivals and Meta 8 (each naming the other); from 2022 both disclose only categories, consistent with counsel advising generic competitive descriptions amid renewed antitrust scrutiny of interlocks. The explicit L2 graph falls from 3 arcs to 1 (N→G), while Nvidia names 18–19 rivals yearly. Without a documented proxy rule, L2 dynamics would reflect legal drafting, not competition. (b) *Semantic drift*: Alphabet adds “AI products and services” (2023), Meta “frontier AI models” (2025), Nvidia Arm-based CPU suppliers (FY2023). (c) *Cross-item redundancy*: Item 1 feeds both L1 and L2, requiring disjoint criteria (*what a firm sells* vs. *whom it claims to face*). (d) *Sticky text*: Item 1A is heavily carried forward [7], so onset dates bound, rather than measure, when a risk became salient.

Preliminary observations (Fig. 1; descriptive, $|V| = 3$). *L1 breadth vs. depth disagree*. Mean J rises from 0.27 to 0.39 as firms enter shared AI and cloud categories, but revenue-weighted R does not converge: $G-M$ falls (0.86→0.77) as Cloud grows within Alphabet, $G-N$ rises only from 0.04 to 0.07, and $M-N$ is 0 throughout because AI is not a separately monetized line at Meta and sits inside “Data Center” at Nvidia. Convergence is thus a category-entry signal, not a revenue-mix signal, and the two weights should be retained as distinct edge attributes. The $M-N$ null overlap coexists with Nvidia’s rising customer concentration (three customers $\geq 10\%$ of revenue in FY2025), pointing to a vertical rather than horizontal tie [8]. *L2*: the proxy rule adds $G \rightarrow N$ from 2023, so the $G-N$ dyad becomes reciprocal only after Alphabet’s AI competitor category appears. *L3*: a dedicated AI risk factor appears for Nvidia, then Alphabet (FY2023), then Meta (FY2024, citing the EU AI Act); an indebtedness risk leaves Nvidia’s filing by FY2025 and enters Alphabet’s in FY2025 after its bond issuance. *Beyond the cohort*: competitor mentions reach 41 external firms, 8 named by ≥ 2 cohort firms (Amazon and Microsoft by all three), marking the boundary for scaling V .

Why the disagreement matters for network science. Binary and weighted projections of the same bipartite firm–category graph encode different processes: J tracks strategic *entry* (a firm starts reporting a category), while R tracks economic *mass* (where revenue actually sits). Entry leads mass when new lines are small, so a widening $J-R$ gap is itself a candidate early signal of structural change. Conversely, results that depend on which projection is used should be flagged as projection-dependent rather than reported as topology. The same logic applies to L2, where explicit and proxy arcs are two projections of one disclosure, and to L3, where theme granularity sets edge density.

Limitations. With $|V| = 3$ every layer is a near-complete triangle, so we report no centrality or community results; the pilot validates rules, not structure. Taxonomy mapping is lossy (“Google Services” bundles advertising, devices, and subscriptions; AI revenue is not separately reported by any firm). Asset geography proxies footprint, not demand, and Nvidia’s FY2025 Singapore revenue reflects centralized invoicing. Coding was done by one annotator; the full study adds a second coder and reports agreement.

Path to scale. An LLM/NER pipeline over S&P 500 10-Ks will use the pilot’s taxonomy, proxy, and alignment rules as annotated ground truth. With $|V| \approx 500$ we will test whether interlayer overlap and breadth–depth divergence grow over time and whether L3 onsets precede L1 entry under both alignments, against degree-preserving layer-shuffled nulls, and add a directed customer–supplier layer. Concretely, we pre-specify three hypotheses: **H1** (breadth leads depth): increases in J_{ij} precede increases in R_{ij} by one or more filing years; **H2** (risk before product): a firm’s adoption of a risk theme precedes its entry into the associated product category; **H3** (disclosure-driven sparsity): the decline in explicit L2 arcs after 2021 is market-wide and concentrated among large platforms, so proxy-injected arcs recover a competition graph whose density is stable over time. Each hypothesis is tested under both alignment conventions and against layer-shuffled nulls, and results are reported separately for explicit and proxy edges.

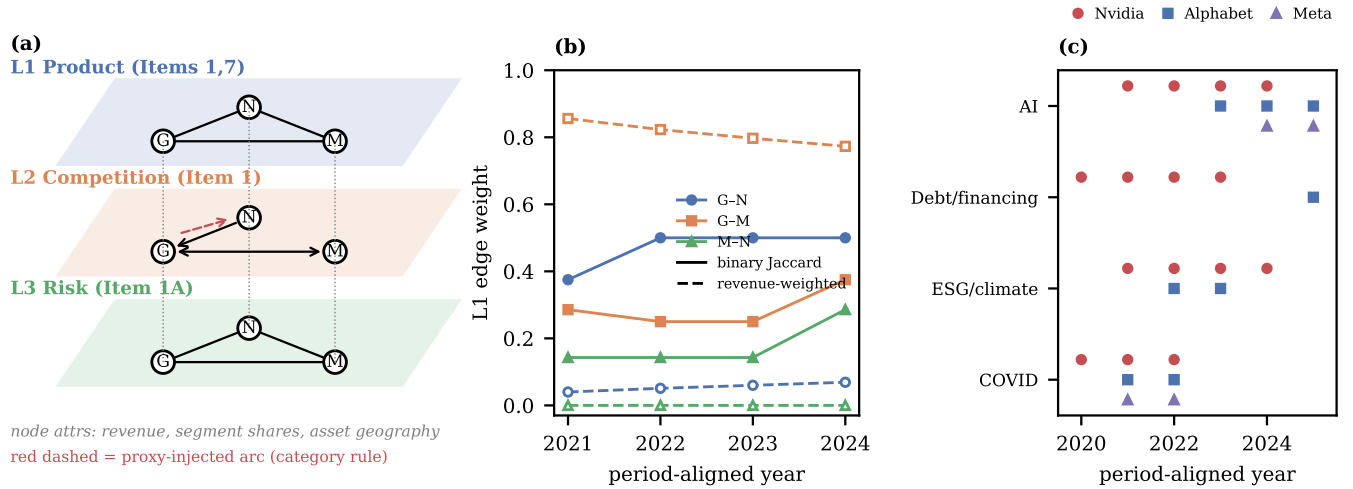


Figure 1: (a) Node-aligned multiplex for G, M, N: L1 product, L2 competition (black: named; red dashed: proxy-injected), L3 risk; Item 8 financials enter as node attributes. (b) L1 edge weights, period-aligned: binary category Jaccard (solid) vs. revenue-weighted Ruzicka (dashed). (c) Disclosure years of selected Item 1A risk themes.

References. [1] M. Kivela et al., *J. Complex Netw.* 2(3):203–271, 2014. [2] S. Boccaletti et al., *Phys. Rep.* 544(1):1–122, 2014. [3] F. Battiston, V. Nicosia, V. Latora, *Phys. Rev. E* 89:032804, 2014. [4] M. De Domenico et al., *Phys. Rev. X* 3:041022, 2013. [5] G. Hoberg, G. Phillips, *J. Polit. Econ.* 124(5):1423–1465, 2016. [6] F. Li, R. Lundholm, M. Minnis, *J. Account. Res.* 51(2):399–436, 2013. [7] L. Cohen, C. Malloy, Q. Nguyen, *Lazy prices*, *J. Finance* 75(3):1371–1415, 2020. [8] L. Cohen, A. Frazzini, *Economic links and predictable returns*, *J. Finance* 63(4):1977–2011, 2008.